

Course Syllabus Part I DSC 550 - Data Mining

3 Credit Hours

Course Description

Data often contains patterns and anomalies that only emerge at large scale. In this course, you will import, clean, manipulate, visualize, analyze, and model structured and unstructured data to extract this information. Model building topics covered include text sentiment analysis, regression, classification, and neural networks. Furthermore, you will learn how to perform feature dimensionality reduction and tune model hyperparameters. The knowledge learned in this course culminates in a term project.

Course Prerequisites

Recommended: DSC 540 Data Preparation

Course Objectives

Students who successfully complete this course should be able to:

1. Import, analyze, and visualize large data sets
 2. Transform data and reduce data dimensionality in preparation for data mining models
 3. Construct natural language processing models
 4. Choose and implement appropriate algorithms for model building
 5. Apply tuning methods to optimize model hyperparameters
 6. Make conclusions based on analysis of data
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Grading Scale

<u>Letter Grade</u>	<u>Percentage Grade</u>	<u>Letter Grade</u>	<u>Percentage Grade</u>
A	≥ 92.5%	C	< 76.5% and ≥ 72.5%
A-	< 92.5% and ≥ 89.5%	C-	< 72.5% and ≥ 69.5%
B+	< 89.5% and ≥ 86.5%	D+	< 69.5% and ≥ 66.5%
B	< 86.5% and ≥ 82.5%	D	< 66.5% and ≥ 62.5%
B-	< 82.5% and ≥ 79.5%	D-	< 62.5% and ≥ 59.5%
C+	< 79.5% and ≥ 76.5%	F	< 59.5%

Topics Outline

- I. Importing and Exploring Data
 - A. Creating data frames
 - B. Exploring, summarizing, and aggregating data
 - C. Dealing with missing data
- II. Data Visualization
 - A. Creating and analyzing data visualizations
 - B. Best data visualization practices
- III. Text, Categorical, and Time/Date Data
 - A. Sentiment analysis
 - B. Preparing text for model building
 - C. Working with categorical data
 - D. Handling dates and times
- IV. Regression models
 - A. Data preparation
 - B. Training, evaluating, and interpreting regression models
- V. Classification Models and Unsupervised Learning
 - A. Data preparation
 - B. Training, evaluating, and interpreting classification models
 - C. Cluster models
- VI. Dimensionality Reduction
 - A. Feature extraction
 - B. Feature selection
 - C. Feature engineering
 - D. Principal component analysis
 - E. Data leakage
- VII. Model Selection and Tuning
 - A. Model selection
 - B. Hyperparameter tuning
 - C. Grid search
 - D. Pipelines
- VIII. Neural Networks
 - A. Artificial neural networks
 - B. Convolutional neural networks
 - C. Image classification
- IX. Implementation
 - A. Creating and organizing a data mining project
 - B. Performing all steps and analysis for a data mining project
 - C. Making a conclusion

This syllabi update reflects grading scale policy updates effective 4/1/2024.